RAM Interface

- Provides access to 1 Gbit of onboard DDR2 RAM.
- Requires modified clocking
  - Design will operate at 37.5 MHz
- Interface consists of IP Core and Wrapper.
  - Provides 8-bit data interface
  - 26-bit addressing
- Single-Step Write Operation
  - Write `data_in` to `address`
- Two-Step Read Operation
  - Request data from `address`
  - Wait for ready
  - Get data & acknowledge

Using the RAM Interface

- General
  - reset: [input] synchronous reset
  - clk: [input] 100MHz clock
  - clkout: [output] 37.5 MHz System Clock
  - sys_clk: [input] 37.5 MHz System Clock (Yes, you have to route this yourself)
  - rdy: [output] '1' signals that the interface is ready for use.
- Writing
  - Single-operation write
  - When \`write_enable\` goes high, the value of \`data_in\` is written to loc \`address\`
- Reading
  - Two-operation write
  - When \`read_request\` goes high, the system queues a READ from loc \`address\`
  - When read is ready, \`rd_data_pres\` goes high, & data is available at \`data_out\`
  - When reading data, acknowledge with a \`1\` on \`read_ack\`